

Co Bao Hieu

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WORK EXPERIENCE

Bosch Global Software Technologies Company Limited

HoChiMinh City, VN

AI Engineer

Nov. 2025 – Present

- Led a small team to design and deliver AI-driven HMI redesign automation and HR AI Interview systems, coordinating frontend, backend, AI service, and deployment workflows.
- Designed and developed, a high-performance, Rust-first AI agent OS and platform encompassing CLI, TUI, web, desktop, including dashboard and Zeroclaw-compatible embedding interfaces support multiples protocols (MCP, A2A, AGP) with many modules, support multiple providers to handle local or remote job. An AI agent level 5 autonomous software engineering AI featuring multi-role Standard Operating Procedures (SOPs), sandboxed execution, and self-correction capabilities. Constructed a semantic Code Built in with Graph engine (codegraph, COG-second-brain, rag) to index and map relationships across large files, and more.
- Architected an end-to-end AI-driven pipeline for Android Automotive OS (AAOSP version 12) to automate HMI UI detect, redesigns (apply multiple resolutions, new theme changes), utilizing a multi-stage UI backend and an LLM-powered LiteLLM Gateway for code generation and APK deployment.
- Developed an end-to-end AI-powered HR platform featuring a semantic matching engine for rapid CV-to-JD shortlisting and an autonomous gRPC video interview pipeline with fraud detection and more.

Software Engineer

Dec. 2021 – Nov. 2025

- Evaluated and ensured compliance for various software licenses (MIT, Apache 2.0, BSD), identifying copyright/copyleft requirements for internal and external applications.
- Scanned and analyzed source code across multiple domains (ATMO, BSH, BCI) to accurately classify components as open-source or commercial software.
- Coordinated with Project and Legal teams via Jira Kanban workflows to issue open-source software certifications and compliance reports.
Developed the EMC AI Internal Web Platform, a comprehensive corporate portal featuring a custom Admin Dashboard CMS for user, post, and permission management.
- Modernized legacy operational tools by transitioning an "Approved Report" web app to a scalable RESTful API architecture and enhancing a ReactJS/Node.js comment generation utility.
- Participated in advanced AI-focused competitions, including Hackathons and the Vehicle Innovation Demonstration Contest, to build edge AI and computer vision prototypes.

Software Engineer Intern

Sep. 2021 – Dec. 2021

- Involved in some newcomer training courses.

DEK Technologies Vietnam

HoChiMinh City, VN

Software Engineer

May 2020 – May 2021

- Knowledge: SIP (RFC 3261, RFC 5359), TCP/IP, UDP, SCTP, VoLTE, TTCN-3, OpenStack, KVM/VMware, Jenkins, UML.
- Executed comprehensive Unit, Characteristic, and Robustness test cases utilizing TTCN-3, meticulously validating OSI Layer 4 protocols (TCP, UDP, SCTP) and VoLTE network architectures.
- Provisioned and customized Linux operating systems integrated with OpenStack and hardware virtualization technologies (KVM, VMware) to construct scalable testing environments.
- Orchestrated continuous integration workflows by managing automated test executions on Jenkins servers within a fast-paced Agile/Scrum framework.
- Authored detailed technical documentation for new features and successfully resolved complex Trouble Requests (TRs) for Ericsson customers, applying deep expertise in SIP standards and UML system design.

Military Insurance Corporation

CanTho City, VN

Insurance Salesman

Aug. 2017 – Dec. 2020

- Consulted with prospective clients to accurately assess their risk profiles, driving sales of customized motorcycle insurance solutions.

MobiFone Service Company Region 9

CanTho City, VN

Support Staff

Apr. 2017 – Feb. 2020

- Formulated business strategies and operational expense plans while overseeing customer mobile data utilization to ensure service efficiency.
- Produced comprehensive daily, quarterly, and annual revenue reports using Microsoft Office, delivering key data visualizations for executive review via internal digital displays and email.
- Spearheaded end-to-end digital marketing campaigns across social media (Facebook, Zalo, YouTube) and e-commerce platforms, driving engagement through custom-created multimedia content.
- Optimized the company's digital presence by authoring SEO-driven content, running targeted Google Ads campaigns, and performing ongoing website maintenance.
- Coordinated traditional marketing events throughout the Mekong Delta region, actively gathering and analyzing participant feedback to refine future promotional initiatives. .

ENVI Technology, LLC

CanTho City, VN

IT Help-desk

May. 2016 – Aug. 2016

- Managed the end-to-end lifecycle of IT assets, ensuring optimal performance and stability for both hardware systems and peripheral devices.

FPT Telecom

CanTho City, VN

Technical Staff

Nov. 2015 – Mar. 2016

- Deployed Fiber-to-the-Home (FTTH) network infrastructure, successfully transitioning legacy copper lines to high-speed optical fiber from exterior plants to internal customer premises.
- Configured and optimized Customer Premises Equipment (CPE), including company-provided Wi-Fi routers, to ensure seamless service delivery and network stability.

PROJECTS

AI Agent OS (Rust, Bun, React, NextJS, Tauri, Sqlite, CodeGraph, RAG, MCP, A2A, AGP, etc.) (R&D)

- **Architected and built a local-first AI agent** platform spanning CLI/TUI, web API, and desktop surfaces from a single 152-crate Rust workspace. Unified four entry points behind one cross-platform command contract, with Zeroclaw-compatible embedding and support for 16 LLM providers (58 models) via MCP, A2A, and agent-to-agent orchestration.
- Designed a layered architecture (shared-commands > agent runtime > providers/tools/MCP) so CLI, web, desktop, dashboard and embeddable surfaces share one business-logic contract without SaaS dependency.
- Built a multi-agent engineering stack (Level 4+ operating posture) with specialist routing, workflow engine, governed tool execution (ToolPolicy, approval flows, audit hooks), and an L5-oriented SWE-agent loop (planner/coder/reviewer/QA with sandboxed execution).
- Implemented a semantic Code Knowledge Graph and RAG pipeline to index, retrieve, and map relationships across large codebases for agent context and codebase navigation.
- Shipped preview/RC-quality installers, release pipelines, and verification gates (make verify, agent benchmark suite scoring L5 on safety/orchestration contracts).

AI HMI Studio Project (Python, Docker, Nodejs, React, NextJS, ViT, YOLOv8, SAM3, GitHub Copilot, GPT-4o, PostgreSQL, Redis, MinIO, FastAPI, Bosch Head Unit, etc.) (R&D)

- Architected an end-to-end AI-driven pipeline to automate the redesign and deployment of Human-Machine Interface (HMI) UIs for Android Automotive OS (AOSP version 12), transitioning from manual to fully autonomous Agile/Scrum workflows.
- Engineered a multi-stage UI processing backend and an LLM-powered Android Engine utilizing a LiteLLM Gateway to dynamically generate production-ready Android code and deployable APKs.
- Implemented an autonomous CI/CD multi-agent system for zero-touch APK builds, bug resolution directly from target Head Unit devices, and complex Git versioning (OEM, AOSP, Web Render).
- Upgraded the platform with advanced interactive tooling, introducing a bidirectional Figma Plugin Engine, a real-time Preview Engine, and a prompt-driven "Grab Editor" for instant UI modifications.

AI HR Interviewer System (Python, NodeJS, React, NextJS, MongoDB, PostgreSQL, GPT-4o, Fast API, etc.)

- Architected an end-to-end AI-powered HR platform using a microservices design (Frontend, Backend, AI Service) behind Traefik/Docker Compose, automating CV screening, interview orchestration, and candidate evaluation.
- Built an LLM-driven CV/JD pipeline that parses unstructured resumes (PDF/DOCX/TXT), deduplicates uploads via content hashing, and runs batch semantic matching against AI-generated job descriptions for rapid shortlisting.
- Designed an autonomous interview workflow that generates tailored MCQ, coding, and behavioral questions through gRPC to the AI service, with MQTT real-time notifications, calendar scheduling, and support for async and live (Teams/WebRTC) sessions.
- Implemented a privacy-oriented evaluation pipeline using FasterWhisper for on-prem speech-to-text, feeding transcripts into LLMs for automated answer scoring and structured HR feedback reports.
- Integrated fraud detection and interview integrity controls — deepfake/liveness checks, code similarity anti-cheating, proctoring logs, and a full audit trail — for both individual and group interview modes.

OSS Internal Web Tools Optimization (Node.js, ReactJS, JavaScript, RESTful API, NoSQL, TailwindCSS, and more)

- **Redesigned and rewrote an internal "OSS Approved Report" web application**, transitioning the system to a modern RESTful API architecture to improve data accessibility and performance.
- **Enhanced an internal comment generation tool** using ReactJS and Node.js to streamline operational workflows and improve cross-team support efficiency.

EMC AI Internal Web Platform (Node.js, ReactJS, JavaScript, RESTful API, NoSQL, TailwindCSS)

- **Developed a comprehensive internal corporate portal** designed to centralize departmental knowledge, featuring dedicated modules for team structures, strategic vision ("About Us"), career pathways, and detailed project repositories.
Engineered a full-featured Admin Dashboard to serve as an internal Content Management System (CMS), enabling administrators to seamlessly manage user accounts, publish dynamic posts, and oversee platform permissions.

Child Presence Detection Module (C++ Migration) (C++, YOLOv8, OpenCV, PyTorch, CMake, Sensor Fusion (Logitech Camera, mmWave Radar))

- **Engineered a sensor fusion algorithm** that combines predictions from an Intel RealSense camera and mmWave radar to accurately classify vehicle occupants (Child vs. Adult) and prevent child-left-behind incidents.
- **Migrated a YOLOv8 object detection model** from Python to a high-performance C++ environment, utilizing CMake for compilation and optimizing execution for edge deployment.
- **Successfully demonstrated the end-to-end alert system** in a live Honda vehicle environment, bridging complex AI models with hardware constraints.

In-Cabin Child Detection System (Vehicle Innovation 2023) (Python, YOLOv5, PyTorch, OpenCV, Raspberry Pi 4, Android)

- **Orchestrated the end-to-end computer vision pipeline**, including collecting, cleaning, and labeling a custom dataset to train a highly accurate YOLOv5 model for in-cabin monitoring.
- **Deployed the AI model onto edge hardware** (Raspberry Pi 4) connected to external webcams for real-time inference within the vehicle.
- **Integrated an automated mobile notification system** to push real-time alerts to an Android device when an unattended child is detected.

Edge AI Driver Monitoring System (Vehicle Innovation 2022) (Temporal Shift Module (TSM), React Native, Edge Computing)

- **Developed an AI-powered driver behavior recognition system** utilizing the Temporal Shift Module (TSM) to detect and classify dangerous actions such as drowsiness, phone usage, and eating.
- **Managed the full machine learning lifecycle**, from raw data collection and multi-class labeling to model training and cross-platform edge deployment on iOS and Android devices.
- **Engineered a closed-loop monitoring architecture** that delivers immediate localized warnings to drivers while simultaneously logging incident events to a centralized cloud server.

SKILLS

Languages: Python, Rust, C++, C, Java, Erlang, JavaScript/TypeScript, Node.js, PHP, C#

Frontend: ReactJS, Next.js, TailwindCSS, HTML/CSS

Backend & API: FastAPI, RESTful API, gRPC, MQTT, LiteLLM Gateway

AI/ML: LLMs, RAG, Code Graph, YOLOv5/v8, SAM3, ViT, FasterWhisper, TSM, OpenCV, PyTorch, and more

Databases & Storage: PostgreSQL, MongoDB, MSSQL, MySQL, Redis, MinIO, SQLite, etc.

DevOps & Tools: Docker, Git, GitHub, GitLab, Gerrit, Jenkins, Jira, Azure Boards, VS Code, Android Studio, Traefik, Alembic, Prometheus, pgAdmin, Conda, PNPM, NPM, Yarn, Makefile, Git.

Platforms: Android Automotive OS, Linux, OpenStack, KVM, VMware, etc.

EDUCATION

Can Tho University <i>Master Computer Science</i>	Dec. 2022
Can Tho University <i>Bachelor Information Technology</i>	Aug. 2018
Can Tho Vocational College <i>Associate of Science Computer Network Management</i>	Sep. 2015